

The schematic diagram illustrates the electrical connections for the PCB assembly. It features two multi-pin connectors, J1 and J2, each labeled "Conn_02x13_Odd_Even".

- J1 Connections:** Pins 1 through 26 are connected to various pins of the XS-LDT-01 module (U1). Specifically, pins 1-10 connect to VDDIO, GPIO7, GPIO6/SPI_S0, GPIO5/SPI_S1, GPIO4/SPI_S2, GPIO3/SPI_S3, GPIO2/SPI_CLK, GPIO3/refCLK, deploy, GPIO0, dump, reset_n, and sleep_n. Pins 11-26 connect to VDDIO, GPIO7, GPIO6/SPI_S0, GPIO5/SPI_S1, GPIO4/SPI_S2, GPIO3/SPI_S3, GPIO2/SPI_CLK, GPIO3/refCLK, deploy, GPIO0, dump, reset_n, and sleep_n.
- J2 Connections:** Pins 1 through 26 are connected to various pins of the XS-LDT-01 module (U1). Specifically, pins 1-10 connect to VDDIO, GPIO7, GPIO6/SPI_S0, GPIO5/SPI_S1, GPIO4/SPI_S2, GPIO3/SPI_S3, GPIO2/SPI_CLK, GPIO3/refCLK, deploy, GPIO0, dump, reset_n, and sleep_n. Pins 11-26 connect to VDDIO, GPIO7, GPIO6/SPI_S0, GPIO5/SPI_S1, GPIO4/SPI_S2, GPIO3/SPI_S3, GPIO2/SPI_CLK, GPIO3/refCLK, deploy, GPIO0, dump, reset_n, and sleep_n.
- Power Supply:** A +3.3V supply is connected to pin 1 of J1 and pin 1 of J2. A +1V1 supply is connected to pin 1 of J2.
- Test Points:** TP1 and TP2 are connected to the output of the U6 TLV74311 voltage regulator. The output of U6 is connected to pin 1 of J2.
- Other Components:** A resistor R9 (3K3) is connected between pins 1 and 2 of J1. A capacitor C2 (1uF) is connected between pins 1 and 2 of J1. A capacitor C6 (1uF) is connected between pins 1 and 2 of J2. A capacitor C10 (1uF) is connected between pins 1 and 2 of J2.

The left diagram shows the FM24W256 chip connected to an Arduino Uno. The chip's VDD is connected to +3.3V, and its VSS is connected to GND. The chip's A0, A1, and A2 pins are connected to the Arduino's A0, A1, and A2 pins, respectively. The chip's SDA and SCL pins are connected to the Arduino's SDA and SCL pins, respectively. The chip's WP pin is connected to the Arduino's WP pin, which is also connected to GND. The chip's CS pin is connected to the Arduino's CS pin, which is also connected to GND. The chip's OE pin is connected to the Arduino's OE pin, which is also connected to GND. The chip's CE pin is connected to the Arduino's CE pin, which is also connected to GND. The chip's I2C address is 0x50.

The right diagram shows the FM24W256 chip connected to an Arduino Uno. The chip's VDD is connected to +3.3V, and its VSS is connected to GND. The chip's A0, A1, and A2 pins are connected to the Arduino's A0, A1, and A2 pins, respectively. The chip's SDA and SCL pins are connected to the Arduino's SDA and SCL pins, respectively. The chip's WP pin is connected to the Arduino's WP pin, which is also connected to GND. The chip's CS pin is connected to the Arduino's CS pin, which is also connected to GND. The chip's OE pin is connected to the Arduino's OE pin, which is also connected to GND. The chip's CE pin is connected to the Arduino's CE pin, which is also connected to GND. The chip's I2C address is 0x50. A note indicates that pin 4 (reset_n) should be pulled low for external I2C read.

The diagram shows the following connections:

- Speaker:** The **LS1 Piezo** output is connected to the **speaker+** and **speaker-** terminals of a speaker symbol.
- Status LED:** The **status_led** signal is connected to the **RUN** pin of a switch symbol. The switch is also connected to a resistor **R4** (2K), which is then connected to **GND**.
- SW1 Switch:** The switch is labeled **SW1** and **SW_DIP x04**. It has four pins on the left (1, 2, 3, 4) connected to **GND** and four pins on the right (8, 7, 6, 5) connected to **sw1**, **sw2**, **sw3**, and **sw4** respectively.

The three circuit diagrams show the connection of the AT25xxx, LM809, and ASE-48Mhz modules to the Raspberry Pi 4. The AT25xxx module is connected to the Pi's VCC, GND, WP, CS, and I2C pins. The LM809 module is connected to the Pi's VCC, GND, RESET, and I2C pins. The ASE-48Mhz module is connected to the Pi's VCC, GND, EN, and I2C pins.

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